

Switching → 3 type memory queue & exercise

Output processing / port

→ buffering required because packets can be lost due to congestion stack or buffer
→ scheduling algorithm → FCFS

RFC → Response

algorithm
→ 3439 rule

RTT → Round Trip Time (250 ms)

HOL → Head of a line Blocking → is an input queue which a queue packet in on input queue must wait for longer though the fabric because it is blocked by another packet at the head of the line the input queue will grow to unbounded length under certain assumption as soon as the packet arrival rate on the input link reaches only 5% of their capacity

Routing Control plane

memory

Bus

Controller



193 is decimal equivalent of first 8 bits
of the address. 32 is the decimal
equivalent of the second 8 bit
address and so on. Inverse 193,
32, 216, 9 is represented in the
binary as 11000001, 00100000,
11011000, 00001001

network interconnecting 3 host interface
and one router interface forms
a separate subnet (of c 450)
a subnet is called an IP network
or simply a network in the Internet.
IP addresser assigns an address to
this subnet in this form 223.1.1.0
/24 where /24 denotes subnet
mask. It indicates the leftmost 24
bits after of the 32 bits address
quantity defines the subnet address.
then the subnet 223.1.1.0/24
consists of 3 host interface within
begins with 1.1, 1.2 & 1.3. So in
general, 223.1.X.X, 16 subnet mask

ICDR $\rightarrow$ Classless Inter Domain
Routing

The Internet assignment address
strategy is known as CIDR. It
generalizes the notion of subnet
addressing. A subnet addressing with
32 bit IP address is divided in
2 parts and again has dotted

193 is decimal equivalent of first 8 bits of the address. 32 is the decimal equivalent of the second 8 bit address and so on. Therefore 193.32.216.9 is represented in the binary as 11000001.00100000.11011000.00001001

network interconnecting 3 host interface and one router interface forms a separate subnet (of C 950) a subnet is called an IP network or simply a network in the Internet. IP addresses assigns an address to the subnet in this form 223.1.1.0/24 where /24 denotes subnet mask. It indicates the leftmost 24 bits after of the 32 bit address. Quantity defines the subnet address then the subnet 223.1.1.0/24 consists of 3 host interface which begins with 1.1, 1.2 & 1.3. So in general, 223.1.X.X, 6 Subnet mask

ECIDR → Classless Inter Domain Routing

The Internet assignment Address strategy is known as CIDR. It uses the notion of subnet and Subnet addressing with addresses is divided in and again has dot-

Step 1 : initialization
 for all $a \neq u$ adjacent to u then
 $D(a) = C_{ua}$

Step 8 : loop

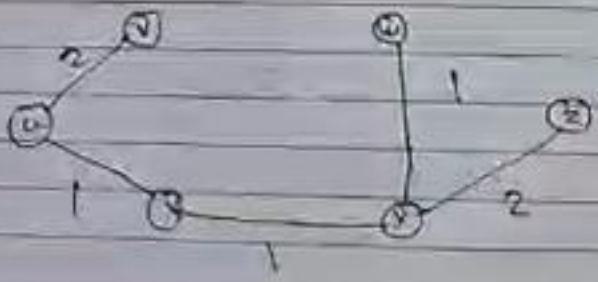
Step 9 : find 'a' not in N'
 such that $D(a)$ is a minimum
 add a to N'

Step 11 : update $D(b)$ for all b
 adjacent to a & not in N'

$$\therefore D(b) = \min(D(b), D(a) + C_{ab})$$

Step	N'	V	W	X	Y	Z
0	u	2, u	5, u	1, u	∞	∞
1	u, x	(2), u	4, x	(1), u	2, x	∞
2	u, x, v	(2), u	(3), x	(1), u	(2), u	∞
3	u, x, v, y	(2), u	(4), x	(1), u	(2), v	∞
4	u, x, v, y, w	(2), u	(3), x	(1), u	(2), x	∞
5	u, x, v, y, w, z	(2), u	(3), x	(1), u	(2), x	(2), y

u x v y w z



Step 1 : initialization
 for all a if adjacent to u then
 $D(a) = (c_u, a)$

Step 8 : loop

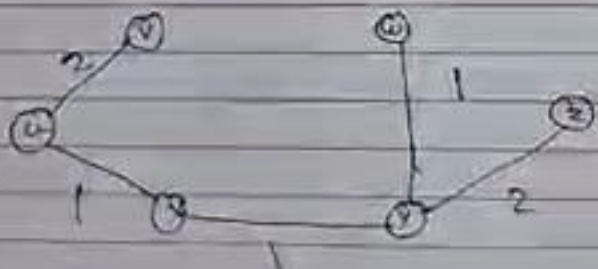
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$$\therefore D(b) = \min (D(b), D(a) + (c_a, b))$$

Step	N'	V	W	X	Y	Z
0	u	2, u	5, u	1, u	∞	∞
1	u, x	2, u	4, x	1, u	2, x	∞
2	u, x, v	2, u	3, x	1, u	2, u	∞
3	u, x, v, y	2, u	3, x	1, u	2, v	4, y
4	u, x, v, y, w	2, u	3, x	1, u	2, x	4, y
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u x v y w z



Step 1 : Initialization
 for all a : if adjacent to u then
 $D(a) = C_{u,a}$

Step 8 : loop

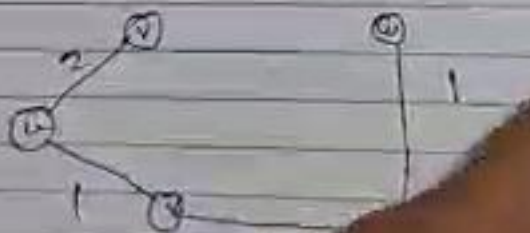
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5	u, x, v, y, w, z	2, u	3, x	1, u	2, x	∞

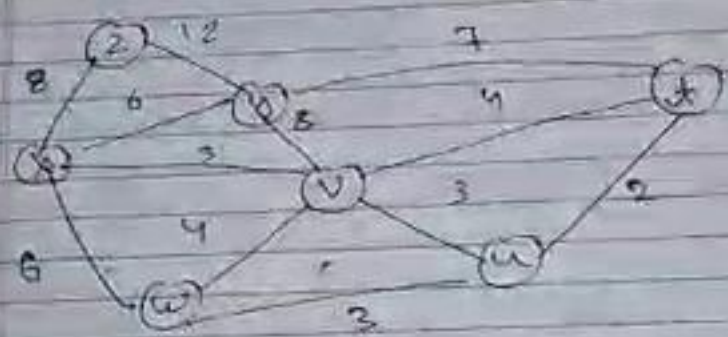
u x v y w z



Destinations

links

V	(u, w)
w	(u, w)
x	(u, w) x
v	(u, w)
z	(u, w)



Distance Vector Routing Algorithm

- Bellman Ford Algo
- Dynamic algo
- Adaptive

let $D_x(y)$: cost of least-cost path from x to y then

$$D_x(y) = \min_v \{ C_{x,v} + D_v(y) \}$$

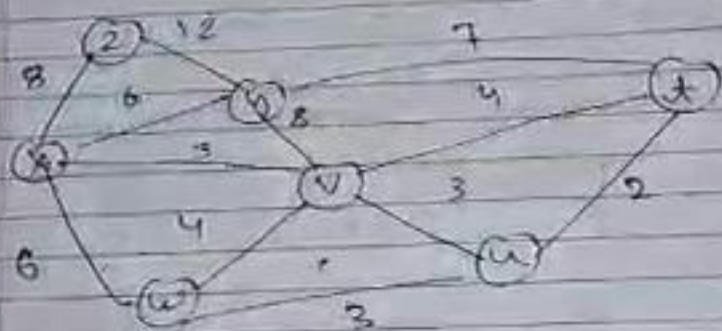
↓
 minimum taken over all neighbors v

distance cost of link from x to v
 + estimated least-cost path cost to y

Destination

links

V	(u, w)
w	(u, w)
x	(u, w) (x)
v	(u, w)
z	(u, w)



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let $D_x(y)$: cost of least-cost path from x to y then

$$D_x(y) = \min_v \{ C_{x,v} + D_v(y) \}$$

↓
minimum taken

over all neighbors v

distance cost
1st estimate



A
C
D

node map
A
B, C
D
D

$$B = 10$$

$$+ 1 - B = 7$$

$$\min(12, 10, 7) = 7$$

Dist ⁿ	Dist ⁿ	Node map
A	7	C, D
B	0	B
C	1	C
D	3	C, D

A	6	D
B	1	B
C	0	C
D	2	D

→ dv

Dest ⁿ	D ⁿ Dist ⁿ	Next Hop
A	4	A
B	8	-
C	2	C
D	0	D

Step 2

A:

Dest ⁿ	Dist ⁿ	Next Hop
A	0	A
B	7	D, C
C	0	D
D	4	D

$$A-B=12$$

$$A-C + C-B = 10$$

$$A-D + D-C + C-B = 7$$

$$\text{Min}(12, 10, 7) = 7$$

B	Dest ⁿ	Dist ⁿ	Next Hop
	A	7	C, D
	B	0	B
	C	1	C
	D	3	C, D

C	Dest ⁿ	Dist ⁿ	Next Hop
	A	6	D
	B	1	B
	C	0	C
	D	2	D